

Each criterion was evaluated against every other using a pairwise comparison method. Values were assigned in 0.25 increments between 0 and 1, where:

- A value of 0.5 indicates equal importance.
- Values of 0.25 or 0.75 indicate one criterion is moderately more important than the other.
- Values of 0 or 1 reflect extreme preference toward one criterion, rendering the other negligible.

The comparisons are symmetric such that the sum of each pair equals 1. Total weights were summed and normalized to create priority values for each criterion.

Criteria	Mass	Continuous Power	Complexity/
Mass		0.25	0.5
Continuous Power	0.75		0.75
Complexity/ Operations	0.5	0.25	
Radiation & Cryo Robustness	0.75	0.5	0.75
Availability/TRL	0.75	0.25	0.75

	Mass	Continuous Power	Complexity/
1	>80 kg	No viable continuous output at Jupiter distance	Continuous
2	65-80 kg	Intermittent; requires massive storage to bridge night	Several
3	50-65 kg	Continuous but significant derating or duty-cycling	A few mechanisms,
4	35-50kg	Continuous with minor load management	Simple one-time
5	<35kg	Constant sun-independent output through full sol incl. night	Passive or nearly

Decision Criteria		Solar Arrays and Li-ion batteries	
Criteria	Weight	Score	Weighted Score
Mass	0.13	2	0.25
Continuous Power	0.28	1	0.28
Complexity/ Operations	0.13	2	0.25
Radiation & Cryo Robustness	0.25	2	0.50
Availability/TRL	0.23	3	0.68
Total Score		1.95	

Radiation & Cryo Robustness	Availability/TRL	Weight	Normalized
0.25	0.25	1.25	0.13
0.5	0.75	2.75	0.28
0.25	0.25	1.25	0.13
	0.5	2.5	0.25
0.5		2.25	0.23
Total		10	

Radiation & Cryo Robustness	Availability/TRL
Fails at 90 K or under >100 krad TID	TRL < 3
Severe output/capacity loss; heavy shielding and	TRL 4-5
Noticeable derating; needs dedicated active	TRL 6-7
Minor derating; standard vault shielding sufficient	TRL 7-8
Essentially immune to radiation and 90 K	TRL 8-9

Analysis of Power Source			
RTG (radioisotope thermoelectric generator)		H2/O2 Fuel cells with tanks	
Score	Weighted Score	Score	Weighted Score
3	0.38	1	0.13
5	1.38	2	0.55
5	0.63	2	0.25
5	1.25	3	0.75
5	1.13	3	0.68
4.75		2.35	